

MUHAMMAD HASSAN SHAHBAZ

AI/ML Engineer | LLM Systems & MLOps | Full Stack Developer

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SUMMARY

Production-focused AI Engineer with experience building RAG systems, multi-agent AI applications, and production ML platforms. Fine-tuned embedding models achieving 93.9% Acc@1, built end-to-end ML pipelines, and contributed to vLLM and LiteLLM open-source projects. Delivered 30+ AI and software projects spanning model training, cost-efficient inference, MLOps, and AWS deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Data Structures & Algorithms, Object-Oriented Programming
LLM / GenAI : LangChain, LangGraph, LlamaIndex, RAG, Agentic AI, Hybrid Search, LoRA/QLoRA, FAISS, Transformers, Prompt Engineering, Vector Databases, BERT, MCP, LLM Evaluation, OpenAI API
ML / DL : PyTorch, Scikit-learn, NLP, Feature Engineering, Model Evaluation, Time-Series Forecasting
MLOps / Cloud: Docker, Kubernetes, Terraform, AWS, MLflow, Prefect, Airflow, CI/CD, GitHub Actions
Backend: FastAPI, Next.js, Node.js REST APIs, WebSockets, Celery, Redis, PostgreSQL, Supabase, , System Design

EXPERIENCE

AI/ML Engineer Intern — CloudQuik Pvt Ltd, Islamabad Jul 2025 – Sep 2025

- Integrated ML models into client-facing product pipelines, reducing manual review steps and cutting processing time by 40%.
- Built a production multilingual (Urdu/English) finance RAG system — fine-tuned embeddings to 93.9% Acc@1 using hybrid BM25+FAISS retrieval, cross-encoder reranking, semantic caching, and confidence gating; embedding model surpassed 400 downloads on Hugging Face.

Freelance AI/ML & Full-Stack Developer — Fiverr (Level 2 · Top 10% Globally) Feb 2024 – Present

- Delivered 30+ projects spanning AI chatbots, SaaS platforms, and automation tools with scalable infrastructure and CI/CD.
- PulmoAI Doctor Assistant:** 10-agent LangGraph diagnostic system (intake → triage → RAG → report) with 3 ML models; 89.8% accuracy on blood-panel disease detection. **GitHub Link:** <https://github.com/Rana-Hassan7272/MedGraph-AI> | [Live Demo](#)

OPEN SOURCE CONTRIBUTIONS & COMPETITIONS

- vLLM (40K+ Github Stars)** | Fixed Anthropic Batch API cost calculation through routing, pricing fallback, JSON translation, test coverage.[\[PR\]](#)
- LiteLLM (25K+ Github Stars)** | Fixed \$0 cost reporting for Anthropic Batch API — root-caused to missing JSON translation layer, added routing fix + unit tests.[\[PR\]](#)
- BirdCLEF+ 2026 · Kaggle (4K+ teams)** | Private LB 0.929 · Best 0.933 — PERCH ONNX + custom SSM + pseudo-label MLP ensemble; ~1 min inference.

Published Articles: [KV-Cache Inference Optimization](#) · [RAG Retrieval-Faithfulness Analysis](#)

PROJECTS

FinGuard AI — Personal Finance Platform (FYP · Live · 🏆 3rd Place, University Final Year Project Competition)) | *FastAPI · LangGraph · BERT · LSTM · Celery · Airflow* | [GitHub](#) | [Live Demo](#)

- Awarded 3rd Place in the University Final Year Project Competition** for FinGuard AI among all final-year CS projects, recognized for innovation, technical depth, and real-world impact.
- Architected a fully automated finance platform ingesting SMS, bank notifications, Gmail, and OCR/PDF into a 22-category ledger via a 7-layer merchant normalizer (MCC → GPS → Knowledge Graph → BERT).
- Trained 3 production ML models: BERT categorization (F1 88.7%), LSTM forecasting (MAE 0.264), Isolation Forest fraud detection (F1 88.7%, AP 95.1%); 91.1% parser pass rate across 6 Pakistani banks.
- Built Finni, a 9-specialist LangGraph advisor (RAG over 69K+ chunks); cached retrieval supported 20 concurrent requests; covers Zakat, Shariah screening, and peer-spending benchmarking.

LLM Systems Engineering — Training, Fine-Tuning & Inference | *PyTorch · Transformers · LoRA/QLoRA · LLM* | [GitHub](#) | [Live Demo 1](#) | [Live Demo 2](#)

- Fine-tuned GPT-style LM with LoRA/QLoRA: 27.7% perplexity reduction, 10× faster inference, lower VRAM usage 88.9% adaptive routing accuracy; 0% failure rate under load across 45-prompt benchmark.
- Benchmarked vLLM (PagedAttention) vs. manual serving: 3.51× throughput at batch=1; 10.25× KV-cache speedup at 1024-token context and adaptive precision routing

MLOps Pipeline | *MLflow · Prefect · Kubernetes · Terraform · AWS · Prometheus · Grafana* | [GitHub](#) | [Live Demo](#)

- Deployed a serverless MLOps pipeline on AWS using Terraform — model training to production in under 10 minutes, hyperparameter tuning, with MLflow tracking, Kubernetes orchestration, and full CI/CD via Github Actions.

EDUCATION

Bachelor of Science In Computer Science — University of Engineering & Technology, Taxila · CGPA: 3.42/4.0 Sep 2022 – Jun 2026

- NSCT 2026 (HEC Pakistan): 92.2 percentile, Top 8% nationwide | LeetCode: 101 solved · 97.2% acceptance · Top 20% globally
- Relevant Coursework: Artificial Intelligence, Artificial Neural Networks, Natural Language Processing, Computer Vision

CERTIFICATIONS

Deep Learning Specialization — [DeepLearning.AI](#) (Coursera) | **AWS Cloud Solutions Architect** — Amazon Web Services (Coursera)
Machine Learning Operations Specialization — Duke University (Coursera) | **Generative AI with LLMs** — [DeepLearning.AI](#) & AWS (Coursera)